

HOW THE MARKS WORK — READ THIS FIRST		
THE STEM LOOKS LIKE	WHAT THE MARKS ARE FOR	FROM HER OWN PAPERS
anything ending “explain your answer” or “justify your answer”	The decision is worth 1. Every other mark is a separate reason. So reasons needed = marks – 1. Write them as separate sentences, not one long one.	“State and justify the stage of mitosis” (4) → name it, then 3 reasons
“State...” or “Identify...” on its own, 1 mark	The name only. No explanation, no sentence. Extra words cannot add marks and cost you time.	“State the model of DNA replication” (1) → semi-conservative
“State and explain...”	Two jobs. The naming is the easy mark — take it even if the explanation defeats you.	“State and explain the role of the following structures” (3)
“Explain the role of X...” 2 marks	What it does, and what that achieves. One without the other is half the marks.	“Explain the role of helicase” (2) → unwinds / breaks H bonds, so that each strand can be copied
“Predict...” or “what would happen if...”	Three beats: what is normally true → what the change breaks → the consequence. Name all three.	“Predict and explain what would happen if histone proteins were negatively charged” (3)
“Deduce...” with a table or figure	Commit to an answer. Quote the number or feature you read off, then reason from it. A hedge scores nothing.	“Deduce if mice X has enhanced OR impaired memory” (4)
“Do you agree?” / “Would X...?”	Say yes or no in the first three words, then justify. Never leave the verdict implied.	“A researcher identifies it as an antiporter. Do you agree?” (2)
“Show your workings clearly”	The arithmetic earns, not only the answer. Write the line even when you can do it in your head.	“10 cells, 5 rounds” (2) → $10 \times 2^5 = 320$
a question about Figure 1	Refer to the figure in your answer — name the labelled letter or the feature you can see. Marks are for reading it, not for reciting the topic.	nearly every structured question is figure-led
“State the consequence at the organ level”	Answer at the level asked. A cell-level answer to an organ-level question scores nothing, however correct.	“...if rough ER is absent” (3) → the gland, not the cell

STUCK? WHAT IS THIS QUESTION ABOUT		
THE QUESTION MENTIONS	IT IS TESTING	GO TO
a protein that is secreted or exported	the secretory pathway	CELLS AND ORGANELLES
ions, charges, or a pump with ATP on it	membrane transport	TRANSPORT
a hormone, a receptor, a distant organ	cell signalling	SIGNALLING
helicase, polymerase, leading/lagging	DNA replication	DNA AND THE CELL CYCLE
operon, repressor, lactose	gene regulation	THE LAC OPERON
codon, tRNA, amino acid chain	translation	DNA TO PROTEIN
ATP, NADH, FADH ₂ , a number	respiration yields	THE ATP TABLE
K _m , rate, substrate concentration	enzyme kinetics	ENZYMES
insulin, glucagon, glycogen	metabolic regulation	HORMONES AND GLUCOSE

CELLS AND ORGANELLES		
THE QUESTION SAYS	WHAT IT IS	THE MARK-EARNING DETAIL
a protein is produced and secreted (casein, milk protein, a hormone)	the secretory pathway, in this order: nucleus → rough ER → vesicle → Golgi → secretory vesicle → plasma membrane → out	name exocytosis for the last step, and name the destination
“role of the nucleus ”	holds the gene; makes the mRNA from it; mRNA leaves by nuclear pores	not “controls the cell” . Tie it to the named protein: the gene FOR CASEIN
“role of the rough ER ”	ribosomes on its surface make secreted and membrane proteins	free ribosomes make the proteins that STAY in the cell
“role of the Golgi ”	modifies , sorts and packages the protein, then buds off the vesicle	“ modify ” is the scoring word , not “transports”
“where are ribosomes made ”	the nucleolus	trap: the answer is not “the nucleus”
“what if the rough ER were absent ”	no secreted protein can be made → no casein → the gland cannot make normal milk	answer at the level asked . An organ question needs an organ answer

CELLS AND ORGANELLES		
THE QUESTION SAYS	WHAT IT IS	THE MARK-EARNING DETAIL
DNA wrapped on a histone	histone is +, the DNA backbone is – (phosphates), so they attract	if histones were negative: they repel → no nucleosome → DNA cannot be packed into the nucleus
“which base pair takes more energy to separate”	G:C	G:C has 3 hydrogen bonds, A:T has 2. More bonds, more energy
1-ring vs 2-ring bases	purines A and G have 2 rings; pyrimidines C and T have 1	a purine always pairs with a pyrimidine
1 NUCLEUSgene → mRNA 2 ROUGH ERribosomes build it 3 GOLGIMODIFIES it, packs it 4 VESICLEcarries it out 5 MEMBRANEexocytosis, out		
No rough ER, no casein, the GLAND fails. Answer at the level asked. Rough ER makes what LEAVES the cell. Free ribosomes make what STAYS.		

TRANSPORT — ACROSS THE MEMBRANE		
THE QUESTION SAYS	WHAT IT IS	THE MARK-EARNING DETAIL
a pump drawing with ATP → ADP on it	primary active transport	energy comes straight from hydrolysis of ATP
the sodium-potassium pump, ions A and B	3 Na⁺ OUT, 2 K⁺ IN , per ATP	read the arrows: whichever leaves is Na ⁺ . Both go against their gradient
“why does it need energy ”	both ions move against their concentration gradient, which cannot happen on its own	then name it: primary active transport , ATP hydrolysis
“is it an antiporter ? do you agree”	Yes . Two different molecules, opposite directions	uniport = one thing symport = two, same way antiport = two, opposite ways
one molecule runs downhill and drags another uphill	secondary active transport (e.g. SGLT1, Na ⁺ with glucose)	the energy is the gradient, not ATP directly
a charged, polar or large molecule at a membrane	cannot cross alone → needs a transport protein	name the barrier: the hydrophobic tails in the middle of the bilayer
the pump is broken by a mutation	gradients are lost → the neuron cannot fire action potentials → signalling between neurons fails	three linked steps, three marks
cell shrinks / cell bursts	shrinks = hypertonic outside bursts = hypotonic outside	always say which way the water moved

SIGNALLING — HORMONES AND RECEPTORS		
THE QUESTION SAYS	WHAT IT IS	THE MARK-EARNING DETAIL
“would this molecule work on that receptor?”	Two gates. Ask both. Gate 1, can it GET IN? hydrophobic yes, hydrophilic no Gate 2, can it FIT? shape, size, charge must match	say WHICH gate fails . Both answers are “no” – the marks are in the reason
testosterone on the estrogen receptor	passes gate 1, fails gate 2	it gets into the cell fine. It is the wrong shape, so no binding, no shape change in the receptor, no response
hydrophilic drug, receptor drawn inside the cell	passes gate 2, fails gate 1	it binds fine in a tube – the question says so. It cannot cross the bilayer, so it never reaches the receptor. Delivery, not binding
made in one organ, acts on a distant one, via blood	endocrine	the tell is distance and the bloodstream
released, acts on neighbouring cells, not via blood	paracrine	into the fluid around it, local targets
needs direct contact between two cells	juxtacrine	the signal is membrane-bound – nothing is released
a cell responds to its own signal	autocrine	

SIGNALLING — HORMONES AND RECEPTORS		
THE QUESTION SAYS	WHAT IT IS	THE MARK-EARNING DETAIL
hydrophobic ligand (steroids, estrogen, testosterone, O ₂)	crosses the bilayer → receptor is inside the cell	
hydrophilic ligand	cannot cross → receptor is on the surface	the ligand's solubility decides where its receptor sits
a tiny signal, an enormous response	signal amplification	happens during transduction , via second messengers cAMP and Ca ²⁺

DNA REPLICATION AND THE CELL CYCLE		
THE QUESTION SAYS	WHAT IT IS	THE MARK-EARNING DETAIL
“state the model of replication”	semi-conservative	one parental strand + one new strand in each daughter. The word alone is the mark
“role of helicase ” (2 marks)	two verbs, one per mark	1 breaks the hydrogen bonds between bases · 2 separates the strands so each can be a template
“more RNA nucleotides on one strand”	the lagging strand	it is built in Okazaki fragments and each fragment needs its own primer . The leading strand needs only one
seals the join between fragments	DNA ligase	makes a phosphodiester bond
removes the primer, fills the gap	DNA polymerase I	pol III builds the strand, pol I erases the primer
“N cells, n rounds, how many?”	N × 2 ⁿ	10 cells, 5 rounds, how many? → $10 \times 2^5 = 320$. write the 2⁵ line – the working is a mark
“parent had 10 chromosomes, the daughter has?”	10	mitosis does not change chromosome number
micrograph: two clusters moving apart , spindle stretched	anaphase	metaphase = one line at the equator anaphase = moving apart telophase = arrived, envelope reforming
“when is the DNA copied?”	S phase , inside interphase	interphase is 90–95% of the cycle. M phase only distributes it

THE LAC OPERON		
THE QUESTION SAYS	WHAT IT IS	THE MARK-EARNING DETAIL
any mutant operon question	run the chain, in this order 1 name the broken part 2 can the repressor still be made? 3 can it bind the operator? 4 can RNA polymerase reach Plac ? 5 so: transcription yes or no	the chain is 4 of the 5 marks . Show it even if you doubt the verdict
compare the figure to the normal diagram first	the exam figure is missing a part on purpose	in the practice paper Pi is absent → no repressor is made → the operator stays free → transcription happens even with no lactose
“which region encodes the repressor ”	lacI	not the operator, not Pi
“mono- or polycistronic ?”	polycistronic	lacZ, lacY, lacA sit under one promoter → one mRNA → several proteins
who binds the promoter vs the operator	promoter → RNA polymerase . Operator → repressor	
Pi vs Plac	Pi serves lacI. Plac serves lacZ, lacY, lacA	break Pi → lose the repressor. Break Plac → lose the enzymes
acetyl on a histone vs methyl on DNA	acetyl = ON, methyl = OFF	HAT adds acetyl (gene on), HDAC removes it. Methyl goes on cytosine and silences
“acetylation rises — does the gene SEQUENCE change?”	No . Only the amount of protein changes	epigenetic = changes activity without changing the DNA sequence

THE ATP TABLE — PER ONE GLUCOSE		
STAGE	WHERE IT HAPPENS	ATP, NADH, FADH ₂
Glycolysis	cytoplasm (no mitochondrion needed)	2 net, 2, 0
Link reaction ×2	mitochondrial matrix	0, 2, 0
pyruvate → acetyl-CoA		releases 2 CO ₂
Krebs cycle ×2 turns	mitochondrial matrix	2 (as GTP), 6, 2
		releases 4 CO ₂
carriers collected		4 direct, 10 NADH, 2 FADH ₂
Oxidative phosphorylation	inner mitochondrial membrane	10 × 3 = 30 2 × 2 = 4 = 34
TOTAL, prokaryote	no mitochondrion, so no delivery cost	4 + 34 = 38
TOTAL, eukaryote	costs 2 ATP to carry glycolysis' NADH into the mitochondrion	38 – 2 = 36
CYTOPLASM GLYCOLYSIS 2 ATP net glucose → 2 pyruvate 2 NADH no oxygen and no mitochondrion needed ↓ MITOCHONDRION MATRIX 2 ATP, 8 NADH link reaction, Krebs 2 FADH₂ ↓ INNER MEMBRANE the chain, ATP synthase 34 ATP		
10 NADH × 3 = 30, and 2 FADH ₂ × 2 = 4.		
4 direct + 34 = 38 , a prokaryote		
36 eukaryote: 2 ATP to ferry NADH in		

RESPIRATION — READING THE QUESTION		
THE QUESTION SAYS	WHAT IT WANTS	ANSWER
human, muscle, liver, yeast , any eukaryote	the 36 total	36 , not 38
bacterium , prokaryote	the 38 total	38
“ net ATP from glycolysis”	4 made, 2 spent priming	2 (4 for two glucose)
“ two molecules of glucose”	double every single number	glycolysis → 4 ATP, 4 NADH
“one turn of the Krebs cycle”	per turn, not per glucose	3 NADH, 1 FADH ₂ , 1 ATP, 2 CO ₂
“CO ₂ in the Krebs cycle , per glucose”	2 turns × 2	4 – the other 2 are from the link reaction
“CO ₂ in total , per glucose”	2 link + 4 Krebs	6 – matches glucose’s 6 carbons
“1 pyruvate via pyruvate dehydrogenase AND the cycle”	naming that enzyme includes the link reaction	4 NADH, 1 FADH ₂ , 1 ATP, 3 CO ₂
“X rounds gave 4 CO ₂ , how much ATP”	2 CO ₂ per round → 2 rounds → 6 NADH	6 × 3 = 18 ATP
“ terminal electron acceptor ” / why oxygen	oxygen , which leaves as water	no O ₂ → NADH is not reoxidised → the whole chain backs up
“what drives ATP synthase ”	protons	electrons move through the CHAIN; protons move through the ENZYME

RESPIRATION — THE PAIRS THAT COST MARKS		
PAIR	THE ONE TELL	BOTH SIDES
aerobic vs anaerobic	is a mitochondrion involved?	aerobic 36/38, to CO ₂ and water anaerobic 2, cytoplasm only
NADH vs FADH₂	where it enters the chain . NADH at the top, so more protons pumped	NADH → 3 ATP FADH ₂ → 2 ATP
ethanol vs lactic acid fermentation	CO ₂ . Ethanol makes it, lactic does not	yeast, froth, bread muscle cramp, yoghurt. Both give 2 ATP
matrix vs inner membrane	Krebs is in the fluid , the chain is in the membrane	protons pile up in the intermembrane space
what fermentation is for	regenerating NAD⁺ , not making ATP	the pyruvate→lactate step makes no ATP. ALL 2 came from glycolysis
liver vs muscle glycogen	only the liver has glucose-6-phosphatase	liver exports glucose to the blood muscle keeps it for itself

DNA TO PROTEIN		
THE QUESTION SAYS	WHAT IT IS	THE MARK-EARNING DETAIL
ribosome + mRNA + tRNAs carrying amino acids	translation	name all three RNAs: mRNA carries codons tRNA carries the amino acid rRNA is the ribosome and makes the bond
read the amino acids off a figure	read the mRNA , left to right, 5' → 3'	do not read the anticodons – they are printed backwards to line up underneath
“the bond joining amino acids ”	peptide bond	nucleotides are joined by PHOSPHODIESTER bonds . Swapping these two scores zero
AUG / UAA, UAG, UGA	start = methionine / the three stops	a stop codon codes for no amino acid
a protein with hydrophobic parts on the OUTSIDE	it will aggregate more when heated	that is the fold inside-out; heat breaks the weak bonds and the exposed hydrophobic patches clump together in water
“which organ for the urea cycle”	the liver	
“the specific cellular location ”	starts in the mitochondria , finishes in the cytoplasm	an organ answer to a sub-cellular question scores nothing
ammonia vs urea	ammonia is toxic , urea is not	NH ₃ + CO ₂ → carbamoyl phosphate, in the mitochondria

ENZYMES — KINETICS AND INHIBITION		
THE QUESTION SAYS	WHAT IT IS	THE MARK-EARNING DETAIL
“compare the catalytic efficiency of two enzymes”	compare their K_m	LOW K_s = HIGH affinity = more efficient . The direction feels backwards, and that is the trap
“what is K _m ”	the substrate concentration at which the rate is half of V _{max}	K _m is a concentration (x axis). V _{max} is a rate (y axis)
read K _m off a table	find V _{max} (the plateau), halve it, read across	plateau 80 → half is 40 → the [S] that gives 40 is K _m
more substrate reverses the inhibition	competitive	same active site. K _m rises, V _{max} unchanged
V _{max} falls , K _m unchanged	non-competitive	binds a different site, so extra substrate cannot displace it
“binds covalently ” / permanent	irreversible	penicillin, aspirin
“enzymes never ...” or “ always ...”	usually false	enzymes DO form transient bonds, and are NOT used up
temperature curves for three organisms	match the peak to the habitat	the options are rarely listed in the same order as the curves

THE SIX ENZYME CLASSES		
IF THE REACTION SHOWS	THE CLASS IS	EXAMPLE
NAD⁺, FAD or NADP⁺ appearing or disappearing; a name ending - dehydrogenase, -oxidase, -reductase	oxidoreductase your deck calls this one “oxidase / dehydrogenase or reductase”	acyl-CoA dehydrogenase, catalase
a group moves from one molecule to another. Any kinase	transferase	hexokinase moves a phosphate
water breaks a bond	hydrolase	lactase, lipase, cellulase
a group added or removed to form a double bond , with nothing broken by water	lyase	pyruvate decarboxylase; enoyl-CoA hydratase adds water across a double bond
one molecule in, one out, same atoms rearranged	isomerase	–mutase, –epimerase
two molecules joined and ATP is spent	ligase	acetyl-CoA carboxylase; any –synthetase

THE SIX ENZYME CLASSES		
IF THE REACTION SHOWS	THE CLASS IS	EXAMPLE
the two tells that settle most of them	water breaking a bond → hydrolase ATP spent joining two things → ligase	anything with NAD or FAD → oxidoreductase

LIPIDS, REDOX AND FREE RADICALS		
THE QUESTION SAYS	WHAT IT IS	THE MARK-EARNING DETAIL
where fatty acid synthesis happens	the cytosol	trap: the pathway begins by exporting acetyl-CoA FROM the mitochondria
where beta-oxidation happens	the mitochondria	build fat in the cytosol, burn it in the mitochondria
one round of beta-oxidation	1 FADH ₂ , 1 NADH, 1 acetyl-CoA	the chain comes out 2 carbons shorter each round
ATP per FADH ₂ / NADH / acetyl-CoA	2 / 3 / 12	your course uses these . Modern books say 1.5 / 2.5 / 10 and would be marked wrong
storage form of lipid, and membrane component	triglyceride , then phospholipid	cholesterol is in the membrane, but its job is fluidity
“unpaired electron”	a free radical , very reactive	it steals an electron from a neighbour, making that one a radical too
“is H ₂ O ₂ a free radical?”	No — but it IS a ROS	no unpaired electron. Radical and ROS are not the same category
“regenerates NADP⁺ from NADPH”	glutathione reductase	not G6PD , which runs the other way. Read the arrow, not the famous name
antioxidants are...	reducing agents	an oxidising agent is itself reduced

HORMONES, GLUCOSE, AND THE THREE GLYCO- WORDS		
THE QUESTION SAYS	WHAT IT IS	THE MARK-EARNING DETAIL
telling the three apart	look for “neo” . If it is there, no glycogen is involved at all	the raw material is amino acids, lactate or glycerol
glycogenesis	making glycogen out of glucose	insulin, after a meal. “–genesis” = birth
glycogenolysis	breaking glycogen back into glucose	glucagon, fasting. “–lysis” = splitting
gluconeogenesis	new glucose from non-carbohydrates	glucagon, starving – only after the glycogen has run out
insulin	glucose is HIGH → brings it down → stores it	activates glycogen synthase, inhibits phosphorylase
glucagon	glucose is LOW → brings it up → releases it	gluca-GONE: glucose is GONE, send more . Activates phosphorylase, inhibits synthase
“which is NOT a gluconeogenesis substrate”	glycogen	it is a carbohydrate. Breaking it down is glycogenolysis
“ketones are made from...”	acetyl-CoA	when it piles up faster than the Krebs cycle can take it
fasting: day 1, day 3, day 30	glycogenolysis, then gluconeogenesis, then ketones	the brain cannot burn fatty acids – they do not cross into it. That is the whole reason ketones exist

BEFORE YOU HAND IT IN	
□Every “explain your answer” — did I give marks – 1 separate reasons?	
□Every “do you agree” — did I actually write yes or no ?	
□Did I answer at the level asked — cell, organ, or sub-cellular?	
□Figure question — did I point at the figure , by label?	
□Eukaryote respiration total is 36 , not 38	
□Two glucose — did I double every number?	
□NADH × 3, FADH ₂ × 2. Never 2.5	
□Low K _m means high affinity	
□“neo” in the word means no glycogen involved	
□Peptide joins amino acids; phosphodiester joins nucleotides	
□Mutant operon — did I write the chain , not just the verdict?	
□Anything blank? Name the structure anyway. Naming is usually the free mark.	